

Sum-Check and GKR Protocols

July 10, 2024

Distributed Lab

 zkdl-camp.github.io

 github.com/ZKDL-Camp



Plan

1 Introduction

2 Multivariate Polynomials. Multilinear Extensions.

Introduction

Recap on Poly-IOPs and NILPs

Almost all protocols we have seen so far work over **univariate** $\mathbb{F}[X]$ **polynomials**. *Typical idea:* we aggregate information into some polynomial: say, $p(X)$ (which is typically a combination of other polynomials), and then check whether $p(u) = 0$ for every $u \in \Omega$:

$$p(u) = 0 \text{ for all } u \in \Omega \iff p(X) = q(X) \prod_{u \in \Omega} (X - u)$$

We check this inequality at a random point $r \leftarrow \$ \mathbb{F}$ and achieve soundness of $1 - \deg p / |\mathbb{F}|$: see **Schwartz-Zippel Lemma**.

For appropriate domains (typically $\Omega = \{\omega^j\}_{j \in [2^r]}$) we reduce all polynomials computations to $O(n \log n)$ complexity.

Motivation

But what if we could work with much smaller degrees?

Sum-Check Protocols

- Instead of n -variate univariate polynomials $\mathbb{F}[X]$ we reduce the problem to $\log n$ -variate multivariate polynomials $\mathbb{F}[X_1, \dots, X_v]$.
- The Schwartz-Zippel Lemma still holds for multivariate case:

$$\Pr_{(r_1, \dots, r_v) \leftarrow \mathbb{S}} [f(r_1, \dots, r_v) = 0] \leq \frac{\deg f}{|\mathbb{S}|}, \quad \mathbb{S} \subseteq \mathbb{F}^v$$

- **Bad news:** divisibility theorems do not hold:

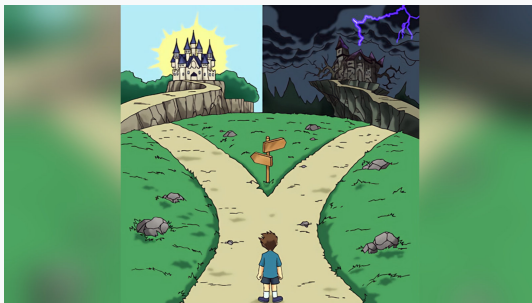
$$f(s_1, \dots, s_v) = 0 \not\iff (X - s_1) \dots (X - s_v) \mid f(X_1, \dots, X_v)$$

Sum-Check meaning

Instead of divisibility checks, we use the **Sum-Check**:

$$\sum_{b_1 \in \{0,1\}} \sum_{b_2 \in \{0,1\}} \dots \sum_{b_v \in \{0,1\}} f(b_1, \dots, b_v) = H$$

Summary in a Nutshell



Univariate World:

$$p(X) = q(X) \prod_{u \in \Omega} (X - u)$$

Multivariate World:

$$\sum_{\mathbf{b} \in \{0,1\}^\ell} f(b_1, \dots, b_\ell) = H$$

Succinct Arguments: Ligerio (2022), Spartan (2020), Libra (2019), Hyrax(2017), GKR (2008), ...

Applications: zkGPT (2025), deep-prove (using *Ceno*) (2024), vSQL (2017), ...

Multivariate Polynomials. Multilinear Extensions.

Some technicalities

Definition (Monomial)

By **monomial** in v variables we call expression $\mu(\mathbf{X}) = X_1^{a_1} \dots X_v^{a_v}$ where $a_1, \dots, a_v$ are non-negative integers. The **degree** of a monomial is naturally defined as $\deg \mu \triangleq a_1 + \dots + a_v$.

Definition (Multivariate Polynomial)

Function $f : \mathbb{F}^v \rightarrow \mathbb{F}$ is called an **v -variate polynomial**, which is denoted by $f \in \mathbb{F}[X_1, \dots, X_v]$, if $f(\mathbf{X})$ is a finite linear combination of v -variate monomials $\{\mu_1(\mathbf{X}), \dots, \mu_n(\mathbf{X})\}$. The **degree** of f is defined as $\deg f \triangleq \max_{i \in \{1, \dots, n\}} \deg \mu_i$.

Example

$f(X_1, X_2, X_3) = X_1^3 + 3X_1^2X_2^2 + X_3^2 + X_3$ is a linear combination of 3-variate monomials $\{X_1^3, X_1^2X_2^2, X_3^2, X_3\}$, thus $f \in \mathbb{F}[X_1, X_2, X_3]$. It has degree $\deg f = 4$, corresponding to the second monomial $X_1^2X_2^2$.

Multilinear Polynomials

Definition (Multilinear Polynomial)

Multivariate polynomial $f \in \mathbb{F}[X_1, \dots, X_v]$ is called **multilinear** if it is linear in each of the variables. Formally we have:

$$f(X_1, \dots, X_v) = a_j X_j + \beta_j, \quad j \in \{1, \dots, n\},$$

where a_j, β_j do not depend on X_j .

Example

For example, $f(X_1, X_2, X_3) = X_1 X_2 + 3X_1 X_3 + X_2 X_3$ is a multilinear polynomial in 3 variables. For instance, for X_1 :

$$f(X_1, X_2, X_3) = (X_2 + 3X_3)X_1 + X_2 X_3.$$

Similarly, $f(X_1, \dots, X_v) = \prod_{i=1}^v X_i$ is a multilinear polynomial.

Boolean Hypercube

Definition (Boolean Hypercube)

By ν -**dimensional boolean hypercube** we simply denote the set $\{0, 1\}^\nu$ (which is a set of binary strings of length ν).

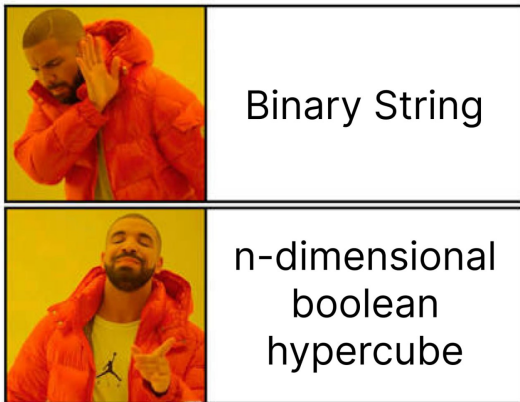


Figure: Cryptographers *love* overcomplicating things.

Multilinear Extensions (MLE)

Definition (Boolean Hypercube Extension)

Suppose we are given the values on the boolean hypercube $f : \{0, 1\}^v \rightarrow \mathbb{F}$. We call $\tilde{f} : \mathbb{F}^v \rightarrow \mathbb{F}$ an **extension** if $\tilde{f}(\mathbf{b}) = f(\mathbf{b})$ for every $\mathbf{b} \in \{0, 1\}^v$.

	0	1
1	2	3
0	1	2

Function $f : \{0, 1\}^2 \rightarrow \mathbb{F}$

Extend 

	0	1	2	3	4
0	1	2	0	0	2
1	2	3	1	1	3
2	0	1	4	4	1
3	0	1	4	4	1
4	2	3	1	1	3

Extension $\tilde{f} : \mathbb{F}_5^2 \rightarrow \mathbb{F}_5$
 $\tilde{f}(X_1, X_2) = X_1^2 + X_2^2 + 1$

Thank you for your attention



zkdl-camp.github.io



github.com/ZKDL-Camp

